

Exploratory Data Analysis Report

Delaney Solubility Dataset

Rows: 1,144

Columns: 5

Missing Values: 0

Duplicate Rows: 15

1. Dataset Overview

Metric	Value
Rows	1,144
Columns	5
Missing Values	0
Duplicate Rows	15
Numeric Variables	5
Categorical Variables	0

2. Dataset Preview - First 10 Rows

MolLogP	MolWt	NumRotatableBond	AromaticProporti	logS
2.5954000000000006	167.85	0.0	0.0	-2.18
2.3765000000000001	133.405	0.0	0.0	-2.0
2.5938	167.85	1.0	0.0	-1.74
2.0289	133.405	1.0	0.0	-1.48
2.9189	187.37500000000003	1.0	0.0	-3.04
1.81	98.96	0.0	0.0	-1.29
1.9352	96.944	0.0	0.0	-1.64
1.4054	118.176	4.0	0.0	-0.43
4.3002	215.894	0.0	0.6	-4.57
2.5654000000000003	132.20599999999996	0.0	0.6	-4.37

3. Descriptive Statistics

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
MolLogP	1144.00	2.45	1.87	-7.57	1.41	2.34	3.41	10.39
MolWt	1144.00	204.63	102.62	16.04	122.13	183.59	270.72	780.95
NumRotatableBonds	1144.00	2.17	2.63	0.00	0.00	1.00	3.00	23.00
AromaticProportion	1144.00	0.36	0.34	0.00	0.00	0.38	0.67	1.00
logS	1144.00	-3.06	2.10	-11.60	-4.33	-2.87	-1.60	1.58

4. Data Types and Variable Information

Column	Data Type	Missing	Unique
MolLogP	float64	0	930
MolWt	float64	0	806
NumRotatableBonds	float64	0	19
AromaticProportion	float64	0	77
logS	float64	0	743

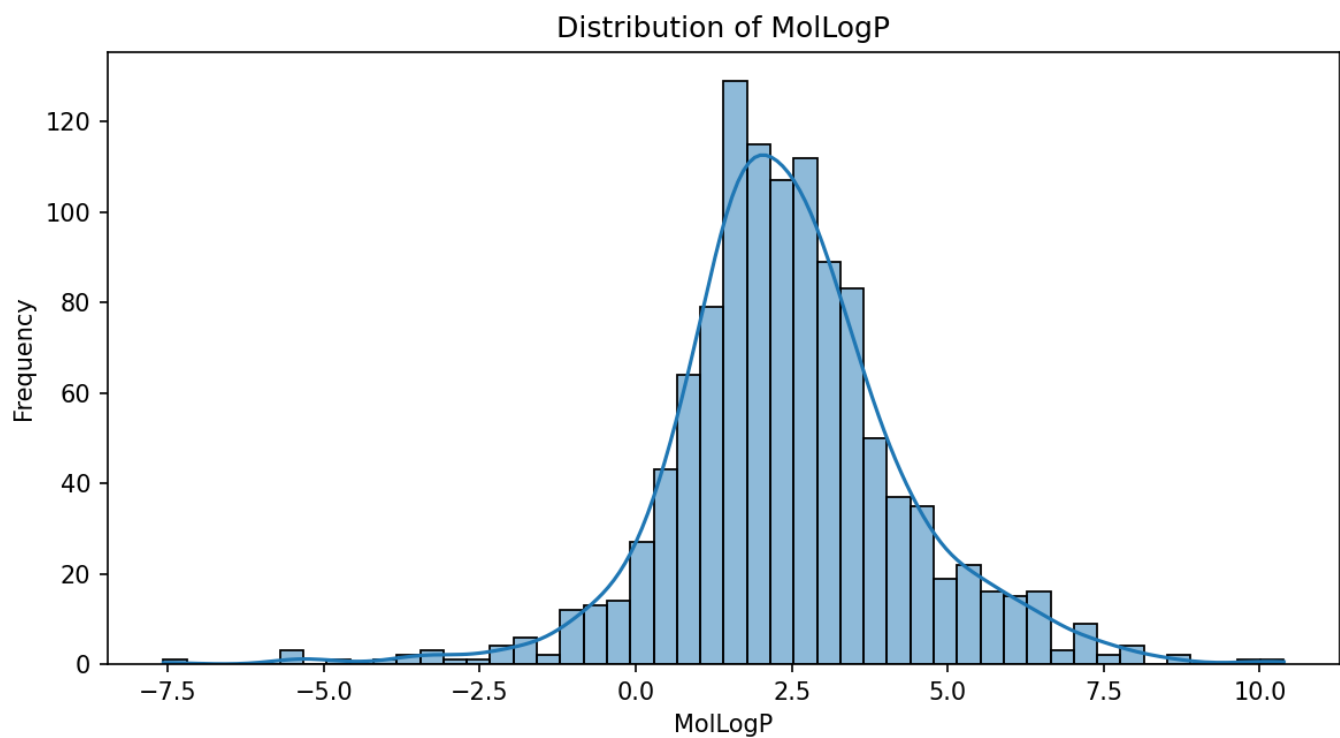
5. YData Profiling Results

Profile Metric	Result
Variables	5
Observations	1144
Missing Cells	0
Duplicate Rows	15

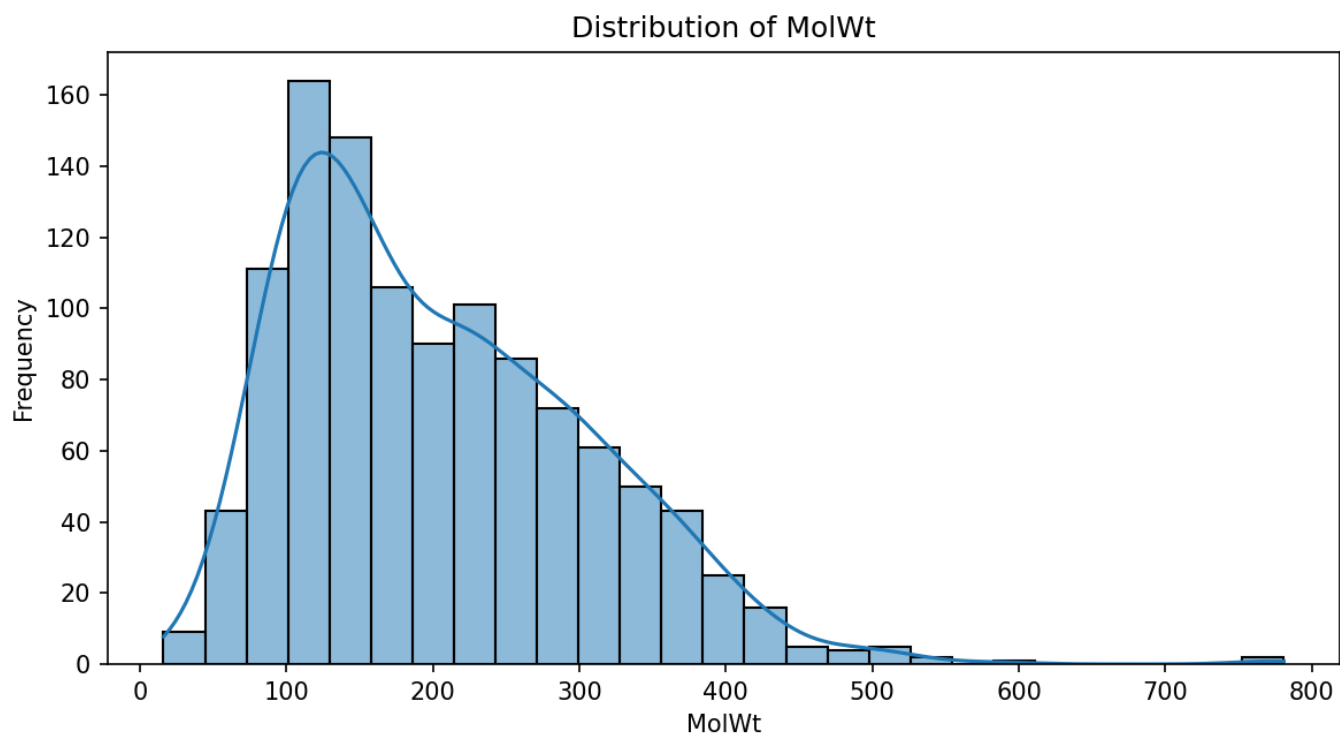
Profiling Alerts

- Duplicate rows detected: 15

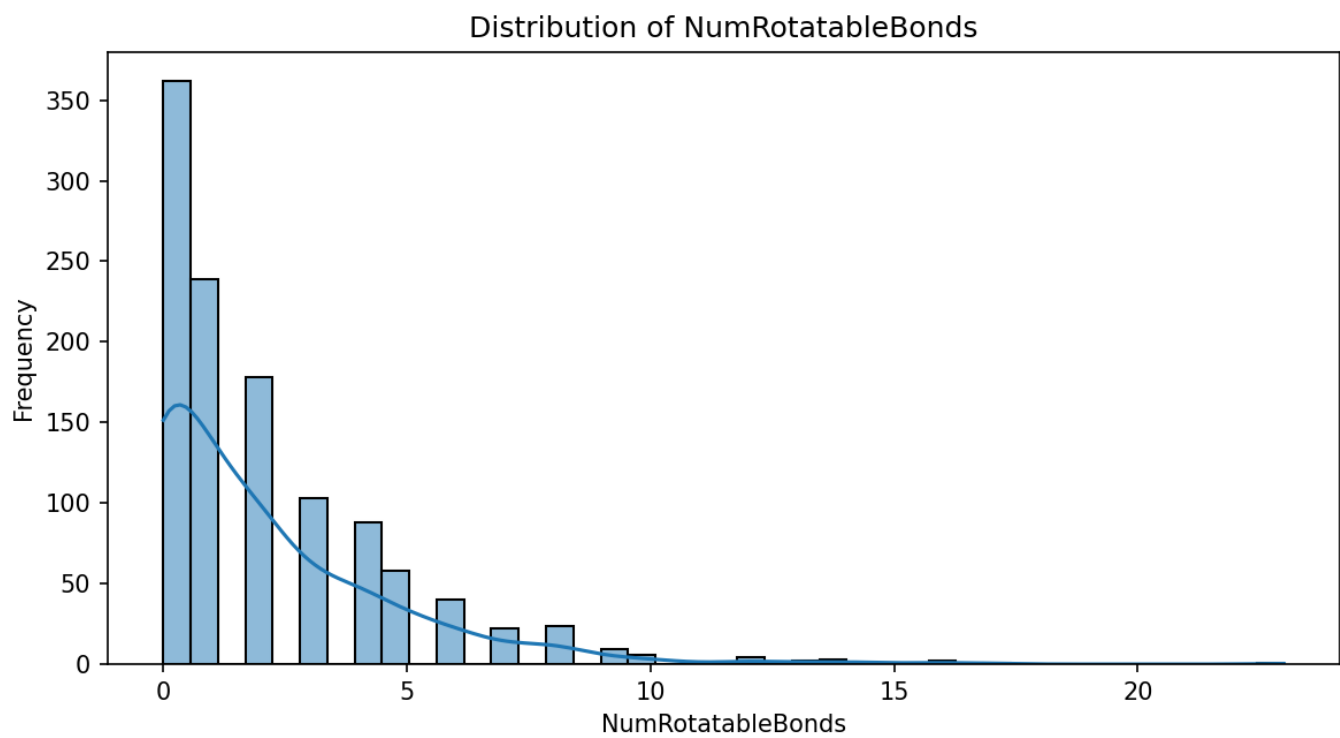
6. Histogram - MolLogP



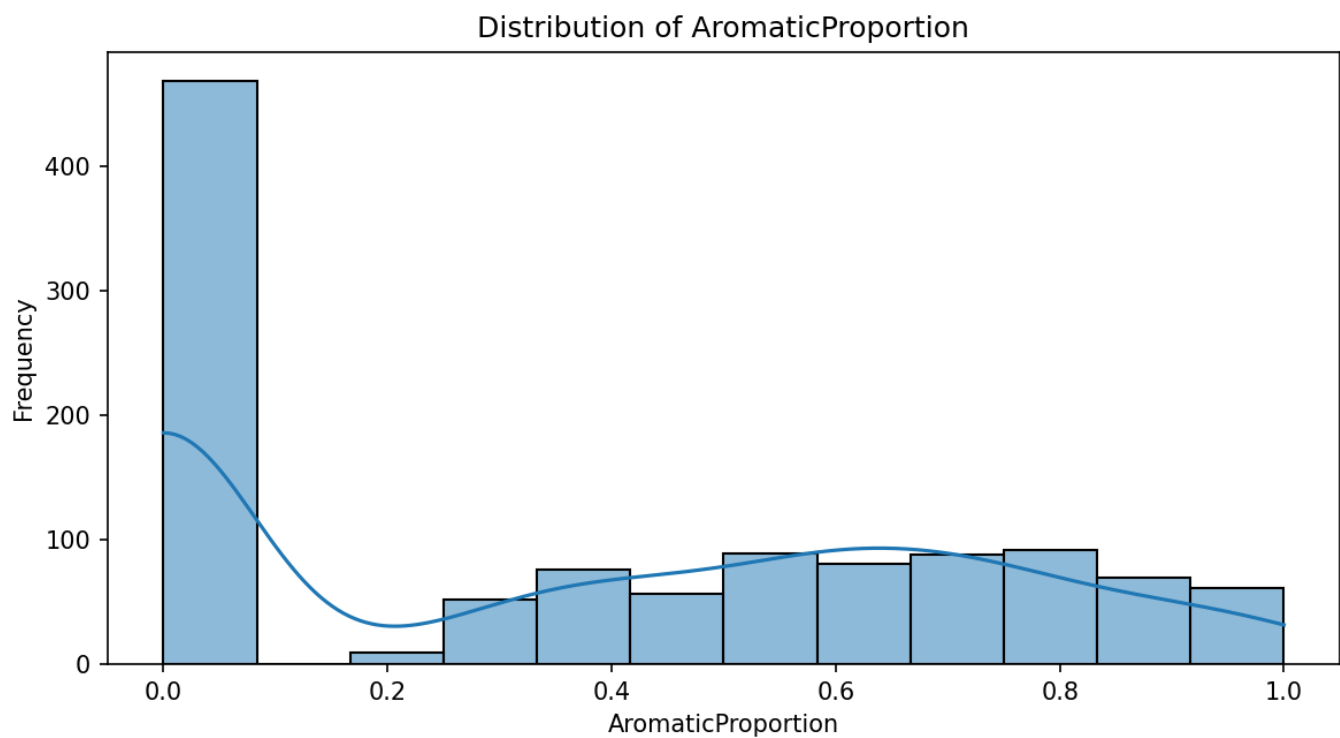
6. Histogram - MolWt



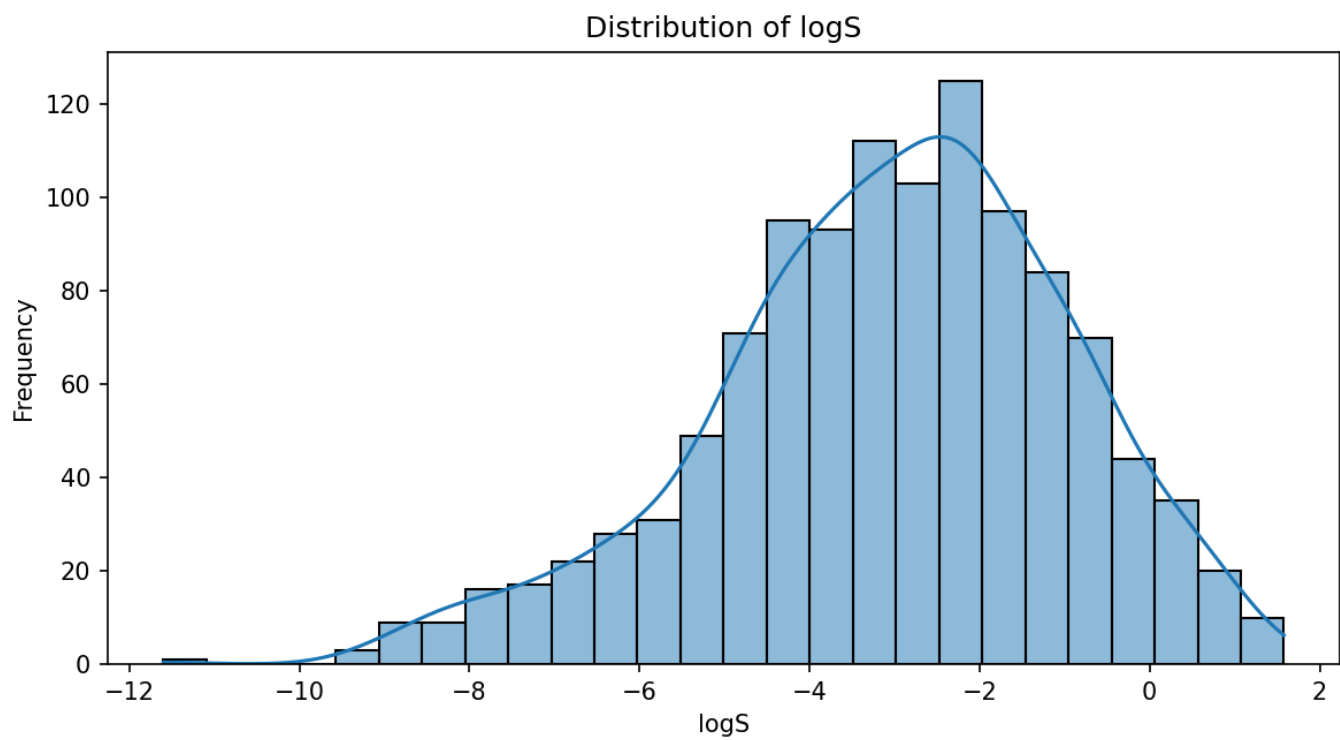
6. Histogram - NumRotatableBonds



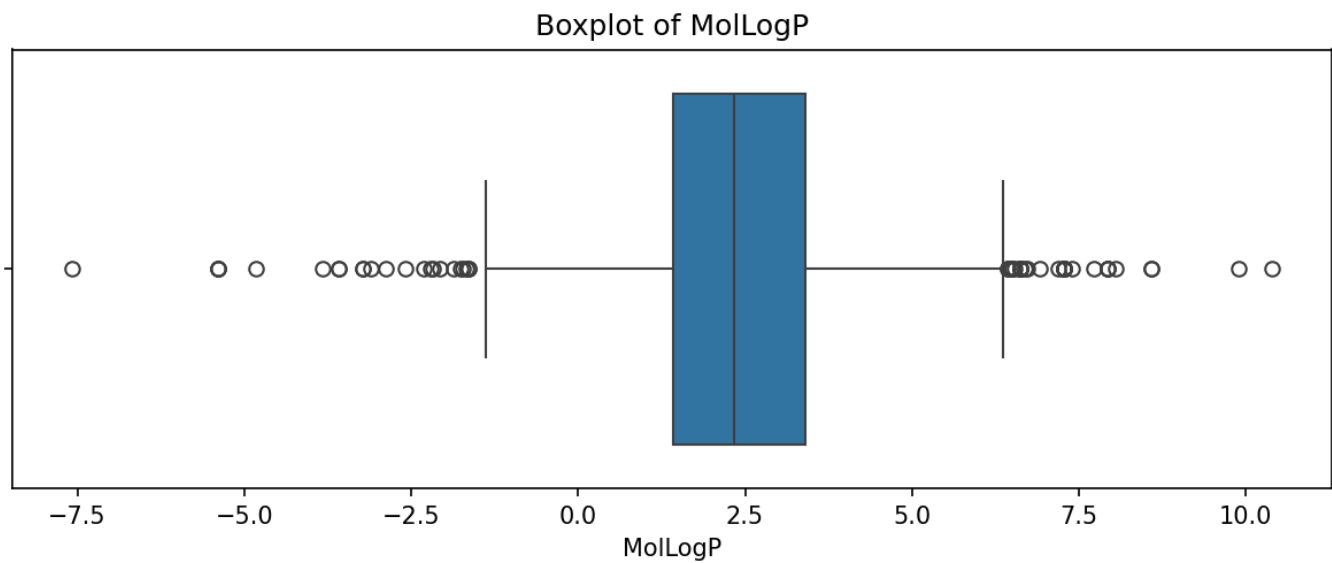
6. Histogram - AromaticProportion



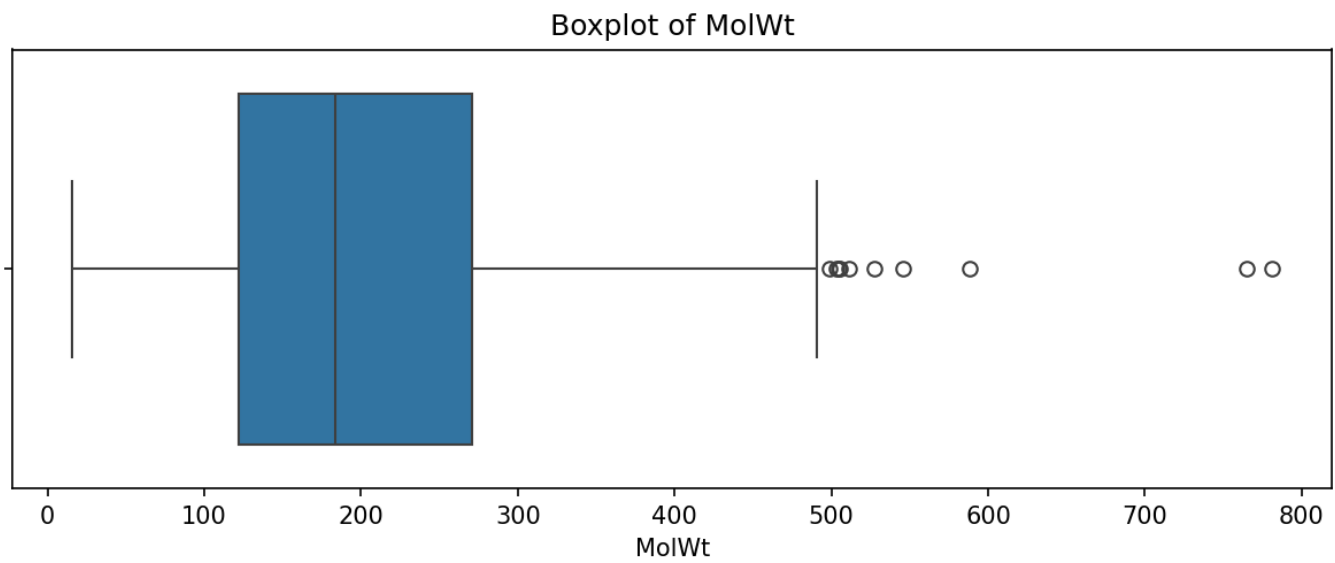
6. Histogram - logS



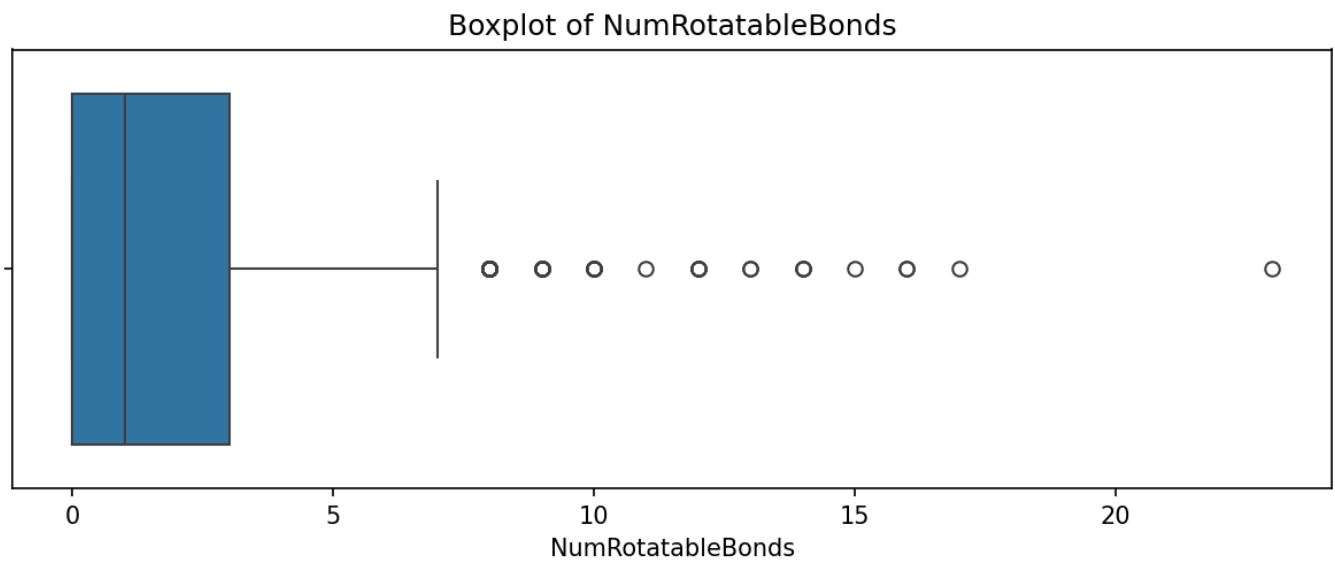
7. Boxplot - MolLogP



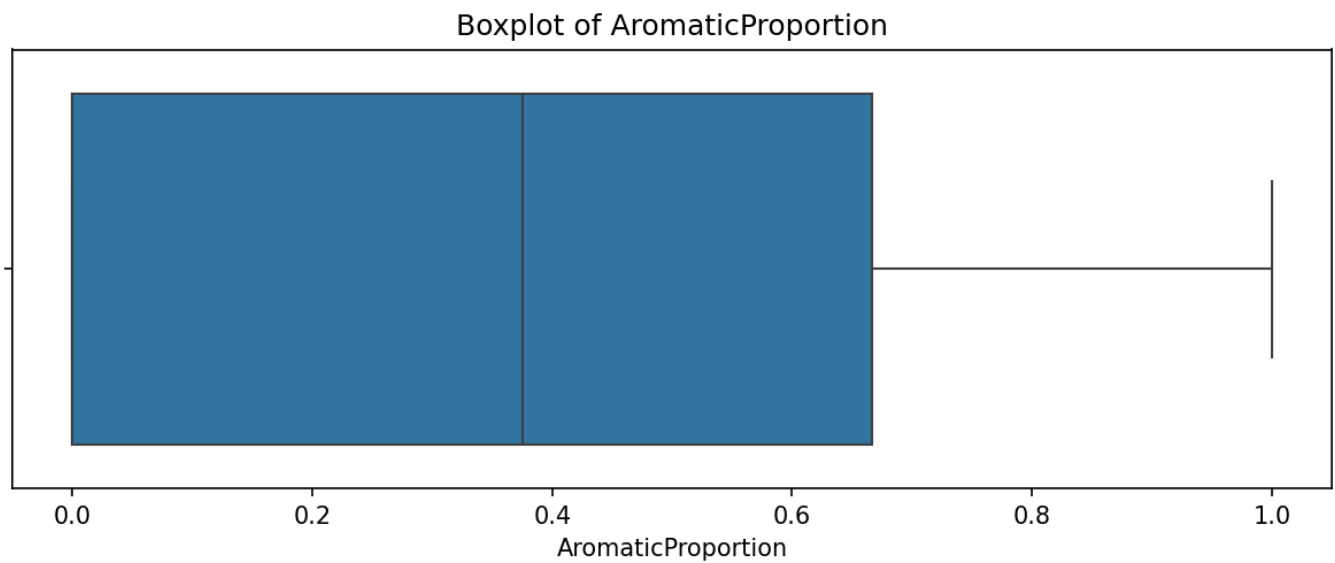
7. Boxplot - MolWt



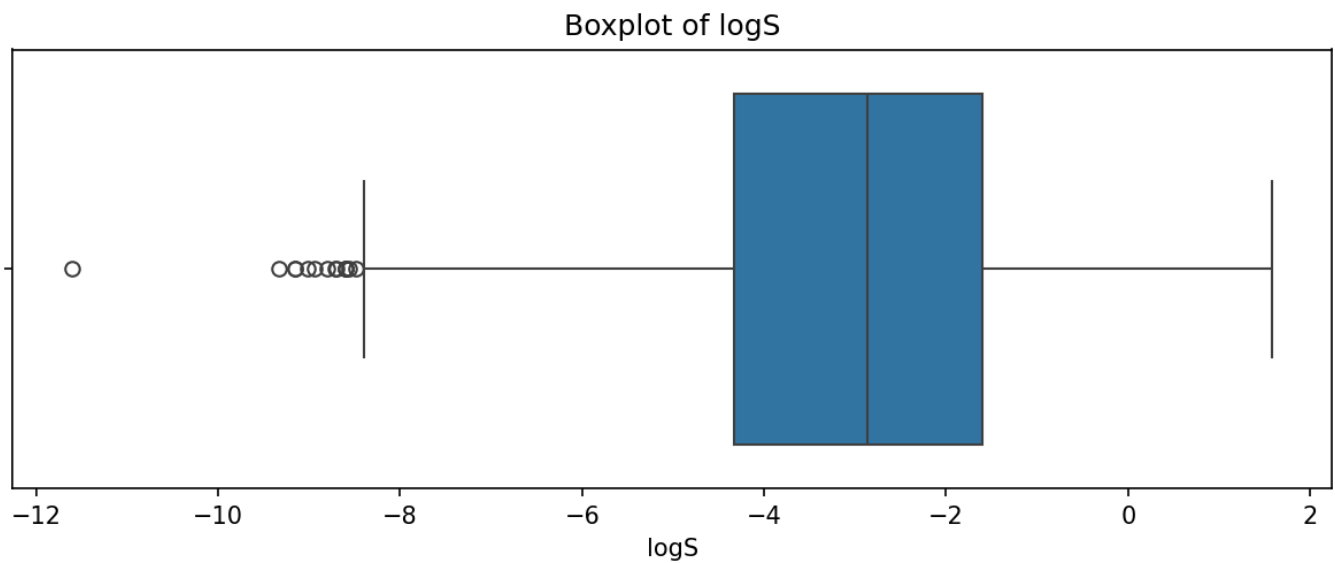
7. Boxplot - NumRotatableBonds



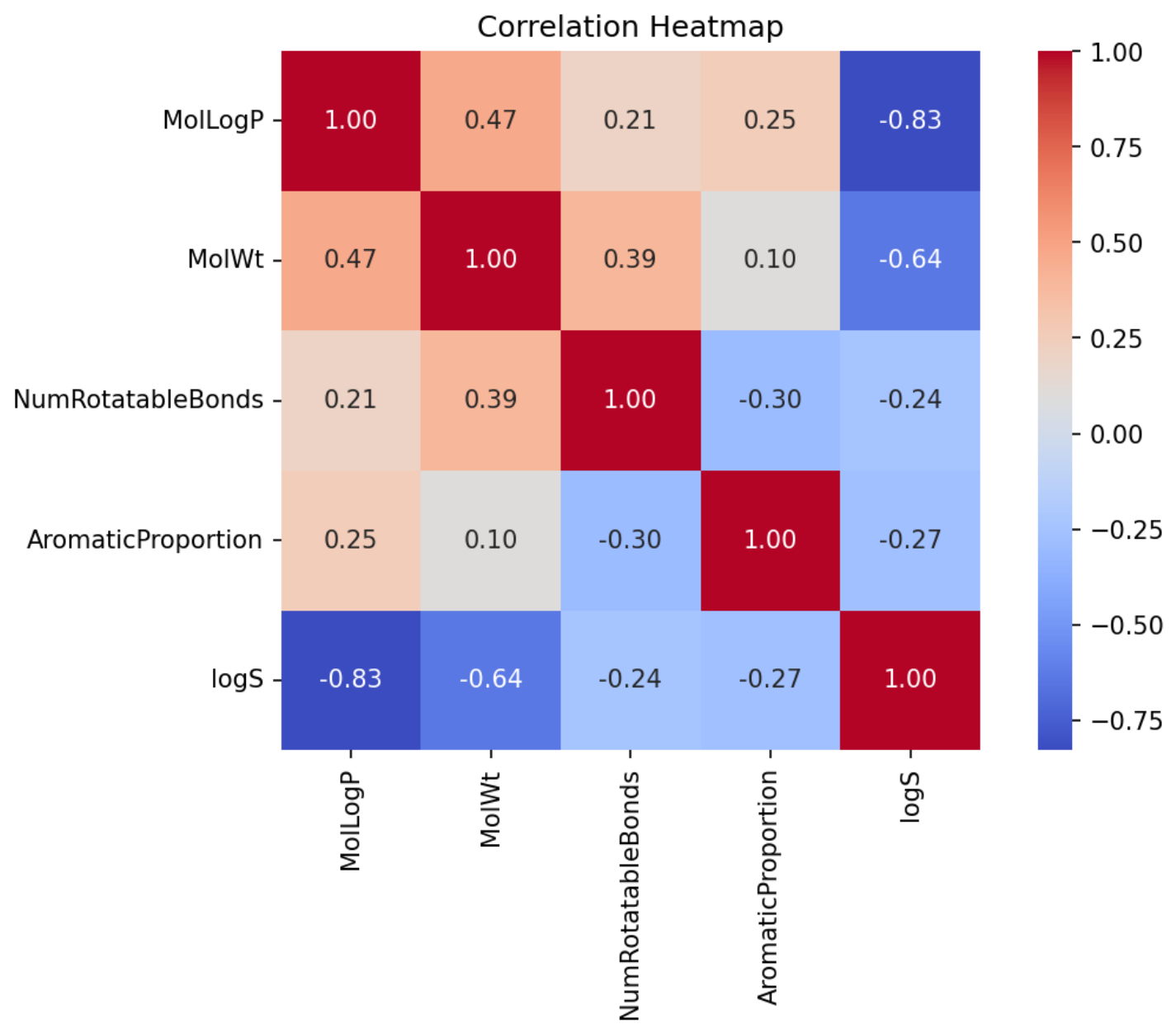
7. Boxplot - AromaticProportion



7. Boxplot - logS



8. Correlation Heatmap



Final EDA Report

This report contains the dataset overview, dataset preview, descriptive statistics, data types, YData profiling summary, distribution plots, boxplots and correlation analysis.